

MARIA NASEEM

AI Solutions Architect | Agentic AI & LLM Systems | Computer Vision | Government & Enterprise Delivery
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PROFESSIONAL SUMMARY

AI Solutions Architect with 10 years of professional experience, including 5+ years architecting and delivering production-grade AI and software systems for enterprise and Pakistani government clients. Architected a facial-recognition Attendance Management System for a provincial government secretariat (designed for thousands of employees) and a YOLOv8-based real-time video analytics solution under the Federal Board of Revenue's Chapter XIV-BA Track & Trace authorization framework — both currently in demonstration / approval phase. Specialise in agentic AI engineering with native MCP server implementations, multi-agent orchestration over OpenAI / Anthropic / Gemini SDKs, RAG with Qdrant / Pinecone, and Human-in-the-Loop (HITL) verification gates. Combine senior technical delivery with proven ministry-level stakeholder engagement (provincial secretaries, ministerial offices, GHQ, SIFC, BOD).

TECHNICAL SKILLS

Agentic AI & LLMs: Multi-Agent Orchestration · MCP Servers (built & deployed) · OpenAI Agents SDK · Anthropic Claude SDK · Google Gemini · LangChain-style tool-use patterns · RAG Pipelines · Embeddings & Similarity Search · Vector DBs (Qdrant, Pinecone) · Pydantic structured outputs · HITL controls · Prompt and Context Engineering

Computer Vision: YOLOv8 / Ultralytics (custom-trained models) · Real-time RTSP video pipelines · Object Detection & Counting · Multi-camera tracking (lap) · Facial Recognition (DeepFace, RetinaFace, YuNet + SFace) · OpenCV · PaddleOCR · FAISS vector similarity

Backend & APIs: Python · FastAPI · async SQLAlchemy 2.x · Pydantic v2 · Celery + Redis · PostgreSQL (psycopg3, asyncpg) · Alembic migrations · WebSockets · JWT/OAuth2 · Microservices Architecture · Node.js

MLOps & Observability: Prometheus · Grafana · Alertmanager · structlog · MLflow concepts · LLM evaluation patterns · A/B testing · model versioning · Pytest + pytest-asyncio · GitHub Actions CI/CD

DevOps & Infrastructure: Docker · Docker Compose · CI/CD Pipelines · Linux · Nginx · Gunicorn · familiarity with Kubernetes patterns

Frontend & Full-Stack: Next.js · React · TypeScript · Tailwind CSS · Streamlit · Playwright (browser automation)

Languages: Python (expert) · SQL · TypeScript / JavaScript · Bash · C++ · PHP

PROFESSIONAL EXPERIENCE

AI Solutions Architect & Full-Stack Developer | Arwen Tech, Islamabad

2025 – Present

- ▶ Architected the technical solution for an **AI/ML-based real-time video analytics and production-counting system** being prepared for vendor authorization under **FBR's Chapter XIV-BA framework** (Sales Tax Rules, 2006) — covering production-line monitoring for Third Schedule goods. Built real-time RTSP video pipelines with YOLOv8 (custom-trained), multi-object tracking (lap), production-line counting, anomaly and stoppage detection, and FastAPI services designed for real-time transmission to a Central Control Unit (CCU). *Currently in demonstration / approval phase.* Stack: Ultralytics, OpenCV, async Postgres, Prometheus telemetry, structured logging.
- ▶ Architected and prepared for deployment a **Facial-Recognition Attendance Management System for a Pakistani provincial government secretariat** (under client confidentiality, pending contract award) — a four-stage CV pipeline (face enrollment → similarity-based candidate retrieval from live RTSP streams → **HITL verification gate** → embedding generation and real-time attendance inference). Multi-camera RTSP ingestion with OpenCV SFace embeddings, FAISS index, threshold-based confidence routing, and a scalable architecture designed for **thousands of government employees**. *Currently in demonstration / approval phase.*
- ▶ Engineered an **agentic notification microservice** — a multi-channel orchestrator (email, WhatsApp, FCM push) using LLM-driven context-aware message generation with deterministic template fallback for guaranteed delivery. Built on FastAPI + Celery + Redis with retry handling, ClamAV virus scanning, and decoupled event-driven processing.
- ▶ Designed and shipped the **Customer Experience Portal** — full SLA lifecycle, contract onboarding, and workflow automation. FastAPI + Next.js + JWT + WebSocket live updates, Celery workers, async Postgres, fastapi-limiter rate limiting.
- ▶ Built a **Document AI / OCR ingestion service** using PaddleOCR + pdfplumber + python-docx with async FastAPI, MIME validation, and audit-grade structured logging.

- ▶ Delivered the **Agentic Observability Platform** — Prometheus → Alertmanager → AI Alert Analyzer (root-cause, severity, next-actions) with self-validating PromQL rules and a native MCP (Model Context Protocol) server for tool integration. Anthropic Claude SDK + FastAPI.
- ▶ Standardised multi-service deployment via Docker Compose and GitHub Actions CI/CD across all delivered microservices.

Executive Coordinator — Corporate & Government Affairs | Digital Operations | Green Tourism Pvt. Ltd. 2024 – 2025

- ▶ Operated as the bridge between the CEO and federal/provincial stakeholder ecosystem: **provincial secretaries and additional secretaries across all four provinces, ministerial offices, GHQ liaison, SIFC (Special Investment Facilitation Council), and the Law Directorate**. Coordinated BOD meetings end-to-end including agenda preparation, MOMs, and follow-through on resolutions.
- ▶ Concurrently led digital operations — corporate web presence, e-commerce, API integrations, and workflow automation.

Co-Founder & Full-Stack Developer | ZUM Services Providers 2023 – 2025

- ▶ Co-founded a multi-services company and served as the primary technical lead, driving architecture, system design, and end-to-end delivery across the full stack (FastAPI, Next.js, PostgreSQL, Docker). Established CI/CD practices using GitHub Actions and automated release pipelines.
- ▶ Designed and delivered solar solution proposals and technical system layouts while leading all digital operations, including website development, marketing initiatives, branding, and digital content creation.

Blockchain Developer | MediaPark 2020 – 2022

- ▶ Designed and delivered scalable native blockchain systems and decentralized applications (DApps), implementing core blockchain architecture including blocks, transactions, consensus validation, and ECDSA-based cryptographic signing for secure authentication and identity protection.
- ▶ Implemented low-level socket-based networking for peer-to-peer (P2P) node communication, enabling distributed synchronization, block propagation, and transaction broadcasting across the blockchain network.
- ▶ Developed compliance-oriented smart contract interaction frameworks with controlled access mechanisms, encryption standards, auditability practices, and structured technical documentation to ensure blockchain security and operational integrity.

Earlier Career (2016 – 2020): Internship at **IBM Pakistan** (2016) — supported PTCL's GPON deployment via PeopleSoft, Siebel, IBM Maximo, and IBM Integration Bus; contributed to GPON training documentation and inventory provisioning. Subsequent operations and IT support roles at Prime Tele Power Solution (2016 – 2017) and Team Work Construction (2018 – 2020).

KEY PROJECTS & OPEN SOURCE

- ▶ **FTE Sales Lead Engine** — Playwright browser automation + AI-driven lead scoring + HITL review layer; reduced manual outreach effort by ~80%.
- ▶ **Agentic Observability Platform** — github.com/Maria-cpp/Agentic-Observability-Platform · Dockerized Prometheus + Grafana + Alertmanager + AI Alert Analyzer with native MCP server.
- ▶ **Bulk Data Import / Document AI** — github.com/Maria-cpp/Bulk-data-import · Async FastAPI + PaddleOCR + Postgres ingestion with validation and audit logging.
- ▶ **Gesture AI Web App** — Real-time hand-gesture detection with Next.js for touchless interaction.

CERTIFICATIONS & EDUCATION

Master of Information Technology — Quaid-e-Azam University, 2016

Agentic AI Level 2 — Professional · PIAIC / Panaversity, 2026 | **Agentic AI Level 1 — Developer Fundamentals** · PIAIC / Panaversity, 2026

FTE AI Program · PIAIC / Panaversity, 2026 | **Generative AI Applications** · Coursera, 2026 | **Build RAG Applications** · IBM / Coursera